Optimization

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Lesson 2

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Example 2

Let B be the following matrix

$$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$

Its **principal submatrices** of order 1 are $[1]$, $[2]$ and $[1]$. Its **principal submatrices** of order 2 are $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$, $\begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. Its **principal submatrix** of order 3 is $\begin{bmatrix} 1 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & -1 & 1 \end{bmatrix}$.

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Example 2

Given that $\det[1] = 1 > 0$, $\det \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = 1 > 0$ and $\det B = 0$, B can not be positive definite. **It is necessary to check the determinants of remaining principal submatrices**.

The determinants of the principal submatrices of order 1 are $\det[2] = 2 > 0$ and $\det[1] = 1 > 0$.

The determinants of the principal submatrices of order 2 are $\det \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} = 3 > 0$ and $\det \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = 0$.

Given that the determinants of all the principal submatrices are non negative and that the determinant of B is null, the matrix is **positive semidefinite**.

Example 3

- Let $C = \begin{bmatrix} -1 & -1 & -1 \\ -1 & -2 & -1 \\ -1 & -1 & -3 \end{bmatrix}$. Given that $\det[-1] = -1 < 0$, $\det \begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix} = 1 > 0$ and $\det C = -2 < 0$, C is **negative definite**.
- Let $D = \begin{bmatrix} -1 & -1 & 1 \\ -1 & -2 & 1 \\ 1 & 1 & -1 \end{bmatrix}$. Given that $\det[-1] = -1 < 0$, $\det \begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix} = 1 > 0$ and $\det D = 0$, D can not be **negative definite**. The determinants of its remaining principal submatrices are $\det[-2] = -2 < 0$, $\det[-1] = -1 < 0$, $\det \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = 0$ and $\det \begin{bmatrix} -2 & 1 \\ 1 & -1 \end{bmatrix} = 1 > 0$. Therefore, D is **negative semidefinite**.

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Operations that Preserve Convexity

- Let $f_1, \ldots, f_m$ be convex functions, $\lambda_1, \ldots, \lambda_m \in \mathbb{R}_+$. Then, $f := \sum_{i=1}^m \lambda_i f_i$ is convex on $\text{dom}(f) := \cap_{i=1}^m \text{dom}(f_i)$.
- Let f be a convex function with $\text{dom}(f) \subseteq \mathbb{R}^d$, $g : \mathbb{R}^m \to \mathbb{R}^d$ an affine function, meaning that $g(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, for some matrix $A \in \mathbb{R}^{d \times m}$ and some vector $\mathbf{b} \in \mathbb{R}^d$. Then the function $f \circ g$ (that maps $\mathbf{x}$ to $f(A\mathbf{x} + \mathbf{b})$) is convex on $\text{dom}(f \circ g) := \{\mathbf{x} \in \mathbb{R}^m : g(\mathbf{x}) \in \text{dom}(f)\}$.

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Local Minima are Global Minima

Lemma

Let $\mathbf{x}^*$ be a local minimum of a convex function $f : \text{dom}(f) \to \mathbb{R}$. Then $\mathbf{x}^*$ is a global minimum, meaning that $f(\mathbf{x}^*) \leq f(\mathbf{y}) \; \forall \; \mathbf{y} \in \text{dom}(f)$.

Proof

Suppose there exists $\mathbf{y} \in \text{dom}(f)$ such that $f(\mathbf{y}) < f(\mathbf{x}^*)$.

Define $\mathbf{y}' := \lambda \mathbf{x}^* + (1 - \lambda)\mathbf{y}$ for $\lambda \in (0, 1)$. From convexity, we get that that $f(\mathbf{y}') < f(\mathbf{x}^*)$. Choosing λ so close to 1 that $||\mathbf{y}' - \mathbf{x}^*|| < \varepsilon$ yields a contradiction to $\mathbf{x}^*$ being a local minimum.

Lemma (Critical Points are Global Minima)

Suppose that f is convex and differentiable over an open domain $\text{dom}(f)$. Let $\mathbf{x} \in \text{dom}(f)$. If $\nabla f(\mathbf{x}) = 0$ (**critical point**), then $\mathbf{x}$ is a global minimum.

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Strictly Convex Functions

Definition

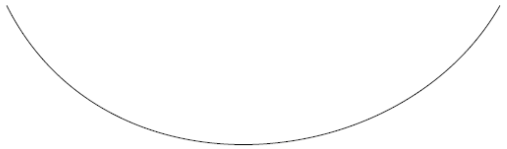
A function $f : dom(f) \to \mathbb{R}$ is strictly convex if

- *$dom(f)$ is convex;*
- *for all $\mathbf{x} \neq \mathbf{y} \in dom(f)$ and all $\lambda \in (0, 1)$, we have:*

$$f(\lambda\mathbf{x} + (1 - \lambda)\mathbf{y}) < \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}).$$



convex, but not strictly convex



strictly convex

Lemma

Let $f : \text{dom}(f) \to \mathbb{R}$ be strictly convex. Then f has at most one global minimum.

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Constrained Minimization

Definition

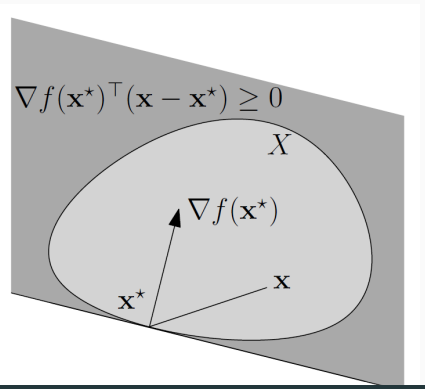
Let $f : dom(f) \to \mathbb{R}$ be convex and let $X \subseteq dom(f)$ be a convex set. A point $\mathbf{x} \in X$ is a **minimizer** *of f over X if*

$$f(\mathbf{x}) \leq f(\mathbf{y}) \quad \forall\, \mathbf{y} \in X.$$

Lemma

Suppose that $f : \text{dom}(f) \to \mathbb{R}$ is convex and differentiable over an open domain $\text{dom}(f) \subseteq \mathbb{R}^d$, and let $X \subseteq \text{dom}(f)$ be a convex set. Point $\mathbf{x}^* \in X$ is a minimizer of f over X if and only if

$$\nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) \geq 0 \quad \forall\, \mathbf{x} \in X.$$



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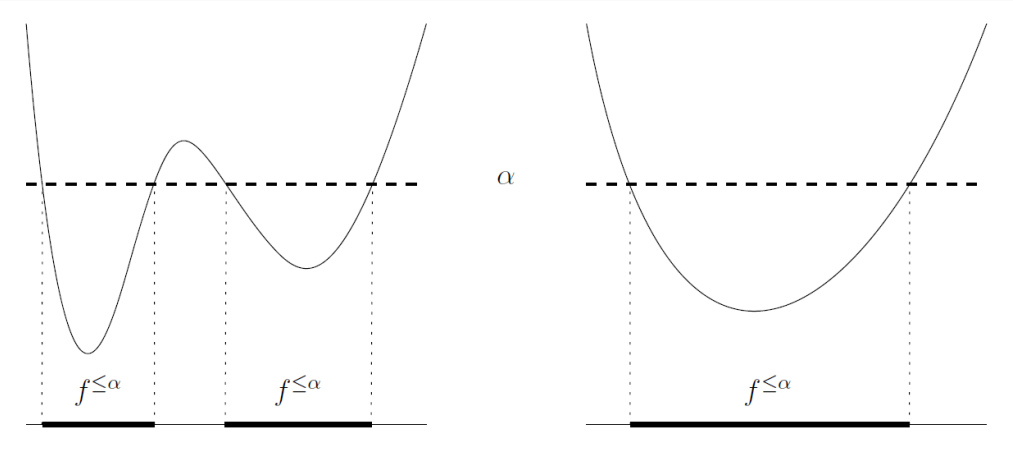
Existence of a Minimizer

How do we know that a global minimum exists?

Not necessarily the case, even if f bounded from below ($f(x) = e^x$).

Definition

$f : \mathbb{R}^d \to \mathbb{R}, \alpha \in \mathbb{R}$. *The set* $f^{\leq \alpha} := \{\mathbf{x} \in \mathbb{R}^d : f(\mathbf{x}) \leq \alpha\}$ *is the* α*-sublevel set of* f.



The Weierstrass Theorem

Theorem

Let $f : \mathbb{R}^d \to \mathbb{R}$ be a convex function, and suppose there is a nonempty and bounded sublevel set $f^{\leq \alpha}$. Then f has a global minimum.

Proof

We know that f (as a continuous function) attains a minimum over the closed and bounded (= compact) set $f^{\leq \alpha}$ at some $\mathbf{x}^*$. This $\mathbf{x}^*$ is also a global minimum as it has value $f(\mathbf{x}^*) \leq \alpha$ while any $\mathbf{x} \notin f^{\leq \alpha}$ has value $f(\mathbf{x}) > \alpha \geq f(\mathbf{x}^*)$.

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Definition

A **Quadratic Program** (**QP**) is an optimization problem whose objective is to minimize or maximize a quadratic function subject to a finite set of linear equality and inequality constraints. By flipping signs if necessary, a quadratic program can be written in the generic form:

$$\begin{aligned} \min \quad & \tfrac{1}{2}\mathbf{x}^\top Q\mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ s.t. \quad & A\mathbf{x} = \mathbf{b} \\ & D\mathbf{x} \geq \mathbf{d} \end{aligned} \tag{1}$$

for some vectors and matrices $\mathbf{c} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{d} \in \mathbb{R}^p$, $A \in \mathbb{R}^{m \times n}$, $D \in \mathbb{R}^{p \times n}$, $Q \in \mathbb{R}^{n \times n}$.

Quadratic Programming

- Quadratic programming models arise in a **variety of practical contexts**.
- The popular **Ordinary Least-Squares** (**OLS**) and **Lasso** estimation procedures in linear regression are also quadratic programs.
- Quadratic programs are also often solved as subproblems in the solution of more general **nonlinear optimization problems**.

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Remarks

The constraint set in (1) is **convex** since it is a system of linear inequalities. Also, the objective function of (1) is convex when Q is a positive semidefinite matrix. Throughout this lesson we assume that Q is **symmetric** and **positive semidefinite**. Therefore problem (1) is a **convex program.**

Definition

A QP model is in **standard form** if it is written as follows:

$$\begin{array}{ll} \min & \frac{1}{2}\mathbf{x}^\top Q\mathbf{x} + \mathbf{c}^\top \mathbf{x} \\ s.t. & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}. \end{array} \tag{2}$$

Theorem

Let a convex QP without constraints be given:

$$\min \quad \tfrac{1}{2}\mathbf{x}^\top Q\mathbf{x} + \mathbf{c}^\top \mathbf{x}. \tag{3}$$

Let $\mathbf{c} \in \mathbb{R}^n$, $Q \in \mathbb{R}^{n\times n}$ and assume Q is symmetric and positive semidefinite. If (3) is **bounded**, then it attains its **minimum**. Furthermore, a point $\mathbf{x} \in \mathbb{R}^n$ is an optimal solution to (3) if and only if:

$$Q\mathbf{x} + \mathbf{c} = \mathbf{0}.$$

When:

- Q is **positive definite**, problem (3) has the unique minimizer $\mathbf{x} = -Q^{-1}\mathbf{c}$.
- Q is **positive semidefinite but not positive definite**, the matrix Q is singular and the problem (3) is either unbounded or has multiple solutions.

Example of a Quadratic programming problem in Finance

Portfolio Selection and Asset Allocation

- Consider an investor who has a certain amount of money to be invested in a number of different securities (stocks, bonds, etc.) with random returns.
- For each security $i = 1, \ldots, n$, estimates of its expected return μ_i and variance σ_i^2 are given.
- Furthermore, for any two securities i and j, their correlation coefficient ρ_{ij} is also assumed to be known.
- If we represent the proportion of the total funds invested in security i by x_i, one can compute the expected return and the variance of the resulting portfolio $\mathbf{x} = (x_1, \ldots, x_n)$ as follows:

$$\mathbb{E}[\mathbf{x}] = x_1\mu_1 + x_2\mu_2 + \ldots + x_n\mu_n = \mu^\top \mathbf{x}, \quad \textbf{expected return}$$

and

$$Var[\mathbf{x}] = \sum_{i,j} \rho_{ij}\sigma_i\sigma_j x_i x_j = \mathbf{x}^\top Q\mathbf{x}, \quad \textbf{variance}$$

where $\rho_{ii} \equiv 1$, $Q_{ij} = \rho_{ij}\sigma_i\sigma_j$ and $\mu = (\mu_1, \ldots, \mu_n)$.

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Example of a Quadratic programming problem in Finance

Portfolio Selection and Asset Allocation

- **Markowitz' Portfolio Optimization** problem (also called the **Mean-Variance Optimization (MVO)** problem) can be formulated via the following convex QP problem:

$$\begin{aligned} \min \quad & \mathbf{x}^\top Q \mathbf{x} \\ s.t. \quad & \textstyle\sum_i x_i = 1 \\ & \mu^\top \mathbf{x} \geq R \\ & \mathbf{x} \geq 0 \end{aligned} \tag{MVO}$$

- The objective function corresponds to the **total variance of the portfolio**.
- The first constraint indicates that the proportions x_i **should sum to** 1.
- The second constraint indicates that the expected return is no less than the **target value** R.
- Nonnegativity constraints on x_i are introduced to rule out **short sales**[1].

[1] selling a security that you do not have

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Supervised Learning vs Unsupervised Learning

Definition (Supervised Learning)

Supervised Learning is the task of learning (inferring) a function f that maps input vectors to their corresponding target vectors, by using a dataset containing a given set of pairs of $(input, output)$ samples.

Examples:

- REGRESSION: the output vectors take one or more continuous values.
- CLASSIFICATION: the output vectors take one value of a finite number of discrete categories. Special case: binary classification.

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Definition (Unsupervised Learning)

Unsupervised Learning is the task of learning (inferring) a function f that maps input vectors to their corresponding target vectors, but without any a priori knowledge about the correct mapping.

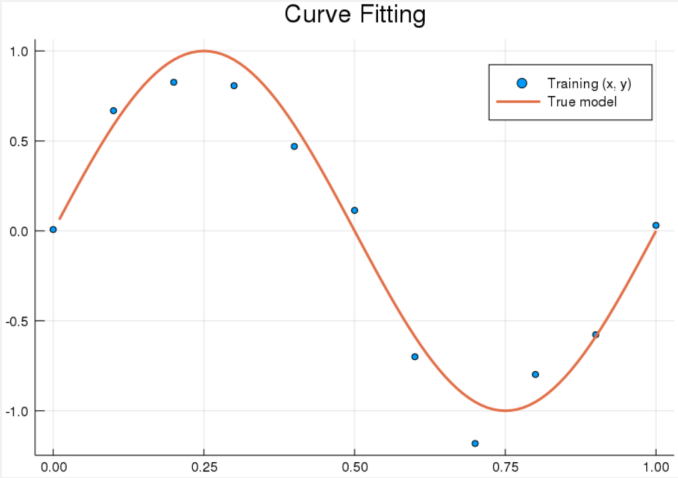
Examples:

- CLUSTERING: The goal of clustering is to group or partition the input vectors (if possible) into k groups or clusters, with the vectors in each group close to each other. In this case, the input vectors represents usually features of objects.
- DENSITY ESTIMATION: The goal is to project the data from a high dimensional space down to two or three dimensions, usually for the purpose of visualization.

Regression Example: Curve Fitting

Example

We are given a training set containing m observations, written $\mathbf{x} = (x_1, ..., x_m)$, together with corresponding observations of the output values, denoted $\mathbf{y} = (y_1, ..., y_m)$. Consider the training set of m data points randomly sampled in the range $[0, 1]$, and the corresponding target values obtained as



$$y_i = \sin(2\pi x_i) + \mathcal{N}(0, 0.02)$$

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Regression Example: Curve Fitting

Learning Model

We want to fit the data using a polynomial function of the form:

$$f(\mathbf{x}; \mathbf{w}) = w_0 + w_1 x^1 + w_2 x^2 + \ldots + w_p x^p = \sum_{i=0}^{p} w_i x^i$$

where p is the order of the polynomial. The polynomial coefficients w_i are collectively denoted by the vector $\mathbf{w}$.

NOTE: While the polynomial function $f(\mathbf{x}; \mathbf{w})$ is a nonlinear function of $\mathbf{x}$, it is a linear function of the coefficients $\mathbf{w}$.

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Regression Example: Curve Fitting

Loss Function

The value of the coefficients will be determined (learned) by fitting the polynomial to the training data. This can be done by minimizing a loss function (or error function) that measures the misfit between the function $f(\mathbf{x}; \mathbf{w})$, for a given value of $\mathbf{w}$, and the training set data points.

A common loss function is the following:

$$L(\mathbf{y}, f(\mathbf{x}; \mathbf{w})) = \frac{1}{2} \sum_{i=1}^{m} (y_i - f(x_i; \mathbf{w}))^2$$

$$L(\mathbf{w}) = \frac{1}{2} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

NOTE: This is the standard sum-of-square error function.

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Regression Example: Quadratic Loss

- As model function $f : X \to Y$ we can start with the following model:

$$y_i = f(x_i) = w_0 + w_1 x_i + \eta_i, \qquad \forall i = 1, \ldots, m.$$

  If m were equal to 2 and $\eta_i = 0$, then we can determine w_0 and w_1.

- Since $m >> 2$ we can only consider the errors in our model

$$\eta_i = y_i - f(x_i) = y_i - w_0 - w_1 x_i, \qquad \forall i = 1, \ldots, m.$$

  and try to **MINIMIZE** them.

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  and try to **MINIMIZE** them.

Regression Example: Quadratic Loss

- In matrix notation we can write

$$A = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ & \dots \\ 1 & x_m \end{bmatrix}, \qquad \mathbf{w} = (w_0, w_1)^\top$$

and then

$$A\mathbf{w} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ & \dots \\ 1 & x_m \end{bmatrix} \cdot \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} w_0 + w_1 x_1 \\ w_0 + w_1 x_2 \\ \dots \\ w_0 + w_1 x_1 \end{bmatrix} = \begin{bmatrix} f(x_1) \\ f(x_2) \\ \dots \\ f(x_m) \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{bmatrix} = \mathbf{y}.$$

And finally

$$\boldsymbol{\eta} = \mathbf{y} - A\mathbf{w}.$$

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Regression Example: Quadratic Loss

- Since we want to minimize a quadratic loss, we have to minimize the following objective function:

$$L(\mathbf{w})^2 = \min \frac{||\boldsymbol{\eta}||^2}{2} = \min \left(\frac{1}{2} \sum_{i=1}^{m} \eta_i^2 \right)$$

$$= \min \left(\frac{1}{2} \sum_{i=1}^{m} (y_i - A^i \mathbf{w})^2 \right)$$

$$= \frac{1}{2} \min (\mathbf{y} - A\mathbf{w})^\top (\mathbf{y} - A\mathbf{w})$$

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Regression Example: Quadratic Loss

- Since we want to minimize a quadratic loss, we have to minimize the following objective function:

$$
\begin{aligned}
L(\mathbf{w})^2 = \min \frac{||\boldsymbol{\eta}||^2}{2} &= \min \left(\frac{1}{2} \sum_{i=1}^{m} \eta_i^2 \right) \\
&= \min \left(\frac{1}{2} \sum_{i=1}^{m} (y_i - A^i \mathbf{w})^2 \right) \\
&= \frac{1}{2} \min (\mathbf{y} - A\mathbf{w})^\top (\mathbf{y} - A\mathbf{w})
\end{aligned}
$$

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Regression Example: Quadratic Loss

- Hence, we have to find

$$\mathbf{w}^* = \arg\min\ (\mathbf{y} - A\mathbf{w})^\top (\mathbf{y} - A\mathbf{w}),$$

where $\mathbf{w} = (w_0, w_1)^\top$ are the unknown.

- Since the problem is quadratic, it is enough to find the point $\mathbf{w}$ s.t.

$$\nabla L(\mathbf{w}) = \mathbf{0}$$

and to check if the associated Hessian is positive definite.

- We have: $L(\mathbf{w}) = \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top A\mathbf{w} + \mathbf{w}^\top A^\top A\mathbf{w}$,

$$\nabla L(\mathbf{w}) = -2\mathbf{y}^\top A + 2A^\top A\mathbf{w} = \mathbf{0},$$

which implies

$$\mathbf{w}^* = (A^\top A)^{-1} A^\top \mathbf{y}.$$

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Regression Example: Regularization

- In order to avoid over-fitting, that is, to obtain weight coefficients with very large weights, the most used techniques is called **regularization**, and involves adding a penalty term to the loss function in order to discourage large coefficients.

  The simplest penalty is the sum of squares of all coefficients:

$$\hat{L}(\mathbf{w}) = \frac{1}{2}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2 + \frac{\lambda}{2}||\mathbf{w}||^2,$$

  where the coefficient λ governs the importance of the regularization term.

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Karush-Kuhn-Tucker (KKT) conditions

Consider the NLP:

$$\begin{aligned} \max \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \ldots, m \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

- How to recognize an optimal solution for a NLP? What are the **necessary** and (perhaps) **sufficient conditions** that such a solution must satisfy? For a **maximization problem:**

Problem	Necessary Conditions	Also Sufficient if
One-variable unconstrained	$\frac{df}{dx} = 0$	$f(x)$ concave
Multivariable unconstrained	$\frac{\partial f}{\partial x_j} = 0,\ j = 1, \ldots, n$	$f(\mathbf{x})$ concave
General constrained problem	Karush–Kuhn–Tucker conditions	$f(\mathbf{x})$ concave and $g_j(\mathbf{x})$ convex

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Karush-Kuhn-Tucker (KKT) conditions

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General constrained problem	Karush–Kuhn–Tucker conditions	$f(\mathbf{x})$ convex and $g_j(\mathbf{x})$ convex

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Karush-Kuhn-Tucker (KKT) conditions

Consider the following NLP **maximization problem:**

$$\begin{aligned} \max \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \ldots, m \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

Its **Lagrangian function** is defined as

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) - \sum_{i=1}^{m} \lambda_i g_i(\mathbf{x}),$$

where $\boldsymbol{\lambda} \in \mathbb{R}_+^m$ is the vector of the **Lagrange multipliers**.

Each λ_i measures the **sensitivity of the objective function** to the corresponding constraints $g_i(\mathbf{x})$.

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Karush-Kuhn-Tucker (KKT) conditions

Consider the following NLP **minimization problem:**

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \ldots, m \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

Its **Lagrangian function** is defined as

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i g_i(\mathbf{x}),$$

where $\boldsymbol{\lambda} \in \mathbb{R}_+^m$ is the vector of the **Lagrange multipliers**.

Each λ_i measures the **sensitivity of the objective function** to the corresponding constraints $g_i(\mathbf{x})$.

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Theorem (Regularity Conditions)

If the vectors $\nabla g_j(\mathbf{x}_0)$ for all $j \in I(\mathbf{x}_0) = \{j \mid g_j(\mathbf{x}_0) = 0\}$ are linearly independent, then the constraints of the problem NLP are qualified.

Example: Two active constraints

Consider two active constraints at $\mathbf{x}_0 = [0, 2]^\top$:

$$g_1(\mathbf{x}) = x_1 + x_2^2 - 1 = 0, \quad g_2(\mathbf{x}) = x_1 - x_2 = 0$$

Their gradients, evaluated in $\mathbf{x}_0$, are $\nabla g_1(\mathbf{x}_0) = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ and $\nabla g_2(\mathbf{x}_0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Let J be the **Jacobian** matrix of the constraints, that is the matrix of the gradients of the active constraints. Then

$$J(\mathbf{x}_0) = \begin{bmatrix} \nabla g_1(\mathbf{x}_0)^\top \\ \nabla g_2(\mathbf{x}_0)^\top \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & -1 \end{bmatrix}.$$

Since $\det(J(\mathbf{x}_0)) = -5 \neq 0$, the gradients are linearly independent, therefore **the constraints are qualified at $\mathbf{x}_0$.**

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Karush-Kuhn-Tucker (KKT) conditions

Theorem (Karush-Kuhn-Tucker conditions)

Let $\mathbf{x}_0$ be a local solution of the constrained maximization problem NLP. If the active constraints in $\mathbf{x}_0$ are qualified, there exists a vector $\boldsymbol{\lambda}^0 \in \mathbb{R}^m$ such that $\boldsymbol{\lambda}^0$ and $\mathbf{x}_0$ are solution of the following system **(KKT conditions)**:

$$\frac{\partial \mathcal{L}(\mathbf{x}_0, \boldsymbol{\lambda}^0)}{\partial x_j} = \frac{\partial f(\mathbf{x}_0)}{\partial x_j} - \sum_{i=1}^{m} \lambda_i^0 \frac{\partial g_i(\mathbf{x}_0)}{\partial x_j} = 0, \qquad j = 1, \ldots, n$$

$$g_i(\mathbf{x}_0) \leq 0, \qquad i = 1, \ldots, m$$

$$\lambda_i^0 \geq 0, \qquad i = 1, \ldots, m$$

$$\lambda_i^0 g_i(\mathbf{x}_0) = 0, \qquad i = 1, \ldots, m$$

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Karush-Kuhn-Tucker (KKT) conditions

Corollary

Assume that $f(\mathbf{x})$ is a concave function and that $g_1(\mathbf{x}), g_2(\mathbf{x}), \ldots, g_m(\mathbf{x})$ are convex functions (i.e., this problem is a **convex programming problem**), where all these functions satisfy the regularity conditions. Then $\mathbf{x}_0$ is an optimal solution **if and only if** all the KKT conditions are satisfied.

- From a practical computational viewpoint, **the method of Lagrange multipliers** and consequently **KKT conditions** are not a particularly powerful procedure. It is often essentially impossible to solve the equations to obtain the critical points.
- Furthermore, even when the points can be obtained, the number of critical points may be so large (often infinite) that it is impractical to attempt to identify a global minimum or maximum. However, for certain types of small problems, this method can sometimes be used successfully.

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Wolfe Dual Problem

Consider the following NLP minimization problem with inequality constraints:

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \ldots, m \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

whose **Lagrangian** function is $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i g_i(\mathbf{x})$ with $\boldsymbol{\lambda} \in \mathbb{R}_+^m$.

We can formulate the **Lagrangian Dual Problem**, that is

$$\begin{aligned} \max_{\boldsymbol{\lambda}} \quad & \theta(\boldsymbol{\lambda}) \\ \text{s.t.} \quad & \lambda_i \geq 0, \quad i = 1, \ldots, m \end{aligned}$$

where the objective function $\theta(\boldsymbol{\lambda}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \inf_{\mathbf{x}} \left(f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i g_i(\mathbf{x}) \right)$ is the **Lagrangian dual function**.

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Wolfe Dual Problem

If f and $g_1 \dots g_m$ are convex and continuously differentiable, then the infimum of $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$ is reached when $\nabla \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$ is equal to $\mathbf{0}$.

It is therefore possible to formulate the **Wolfe Dual Problem (WDP)** as follows

$$\begin{aligned} \max_{\mathbf{x}, \boldsymbol{\lambda}} \quad & f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i g_i(\mathbf{x}) \\ \text{s.t.} \quad & \nabla f(\mathbf{x}) + \sum_{i=1}^{m} \lambda_i \nabla g_i(\mathbf{x}) = \mathbf{0} \qquad (*) \\ & \lambda_i \geq 0, \quad i = 1, \dots, m \end{aligned}$$

where the **KKT stationarity condition** $(*)$ are included as a constraint.

It should be noted that, if f and $g_1 \dots g_m$ are convex, then **strong duality** usually holds, but:

- if the equality constraint is nonlinear, the WDP is a nonconvex problem. Then **strong duality** is not guaranteed;
- however, **weak duality** always holds. This means that the dual objective provides a **lower bound** to the NLP primal minimum.

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Slater's Conditions for Strong Duality

Are there any conditions, beyond convexity, that guarantee strong duality?

Among these conditions, called *constraints qualifications*, there are the **Slater's condition** that guarantee strong duality to hold if f and $g_1 \dots g_m$ are convex.

Let $\mathcal{D}$ be the set on which f and each g_i are defined, that is $\mathcal{D} := dom(f) \cap \mathcal{G}$, where $\mathcal{G} := \bigcap_{i=1}^{m} dom(g_i)$. We say that $g_1, \dots, g_m$ satisfy the **Slater's condition** if

$$g_i(\bar{\mathbf{x}}) < 0, \quad i = 1, \dots, m,$$

for some $\bar{\mathbf{x}}$ in the *relative interior* of $\mathcal{D}$. This is equivalent to state that $\bar{\mathbf{x}}$ is *strictly feasible*, i.e. all m constraints are satisfied with strict inequality.

Theorem (Slater's condition)

If f and $g_1 \dots g_m$ are convex and $g_1 \dots g_m$ satisfy the **Slater's condition**, then **strong duality holds**.

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Slater's Conditions for Strong Duality

The Slater's condition for strong duality can be **refined** if some of the constraint functions $g_1, \ldots, g_m$ are *affine*.

Let $g_1, \ldots, g_k$ be **the first k affine functions**, with $k \leq m$. We say that the convex functions $g_1 \ldots g_m$ satisfy the **refined Slater's condition** if

$$g_i(\bar{\mathbf{x}}) \leq 0, \quad i = 1, \ldots, k \qquad g_i(\bar{\mathbf{x}}) < 0 \quad i = k+1, \ldots, m,$$

for some $\bar{\mathbf{x}}$ in the *relative interior* of $\mathcal{D}$. This condition is equivalent to state that $\bar{\mathbf{x}}$ is *strictly feasible* **only for the nonaffine constraints**, while the affine inequalities do not need to hold with strict inequality.

Remark

Slater's and refined Slater's conditions not only **guarantee strong duality** when f and $g_1 \ldots g_m$ are convex. They also imply that the **dual optimal value** is **attained** when the *dual problem* is bounded from below.

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Example: Least-squares solution of linear equations

We consider the problem

$$\begin{aligned} \min \quad & \mathbf{x}^\top \mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b}, \end{aligned}$$

where $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^p$, and $A \in \mathbb{R}^{p \times n}$. This problem has **no inequality** constraints and p **linear equality** constraints.

Its Lagrangian function is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{x}^\top \mathbf{x} + \boldsymbol{\lambda}^\top (A\mathbf{x} - \mathbf{b}),$$

which is a **convex quadratic** function of $\mathbf{x}$, with $\boldsymbol{\lambda} \in \mathbb{R}^p$.

As mentioned before for the WDP, the **dual function** of $\mathcal{L}(\mathbf{x}, \lambda)$ is

$$\theta(\boldsymbol{\lambda}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \inf_{\mathbf{x}} \left(\mathbf{x}^\top \mathbf{x} + \boldsymbol{\lambda}^\top (A\mathbf{x} - \mathbf{b}) \right).$$

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Slater's Conditions for Strong Duality

The **convexity** of $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$ implies that the minimizing $\mathbf{x}^*$ can be found from the **optimality condition** $(*)$, that is

$$\nabla \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = 2\mathbf{x} + A^\top \boldsymbol{\lambda} = \mathbf{0},$$

which yields to $\mathbf{x}^* = -\frac{1}{2} A^\top \boldsymbol{\lambda}$.

Then, the dual function can be written as

$$\begin{aligned}\theta(\boldsymbol{\lambda}) =& \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}) = \left(-\frac{1}{2} A^\top \boldsymbol{\lambda} \right)^\top \left(-\frac{1}{2} A^\top \boldsymbol{\lambda} \right) + \boldsymbol{\lambda}^\top \left(A\Big(-\frac{1}{2} A^\top \boldsymbol{\lambda} \Big) - \mathbf{b} \right) = \\ &= -\frac{1}{4} \boldsymbol{\lambda}^\top A A^\top \boldsymbol{\lambda} - \mathbf{b}^\top \boldsymbol{\lambda}.\end{aligned}$$

which is a **concave quadratic** function for which **weak duality** holds.

Therefore, $\theta(\boldsymbol{\lambda})$ provides a **lower bound** for the primal problem, that is

$$-\frac{1}{4} \boldsymbol{\lambda}^\top A A^\top \boldsymbol{\lambda} - \mathbf{b}^\top \boldsymbol{\lambda} \leq \inf\{\mathbf{x}^\top \mathbf{x} | A\mathbf{x} = \mathbf{b}\}.$$

What about the **strong duality**?

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Slater's Conditions for Strong Duality

Starting from the **dual problem**, formulated as

$$\max_{\boldsymbol{\lambda}} \theta(\boldsymbol{\lambda}) = \max_{\boldsymbol{\lambda}} \Big(-\frac{1}{4}\boldsymbol{\lambda}^\top A A^\top \boldsymbol{\lambda} - \mathbf{b}^\top \boldsymbol{\lambda} \Big),$$

we note it is an **unconstrained concave maximization problem**.

This guarantees that the Slater's condition is always satisfied if the **primal problem is feasible**.

Therefore, the **strong duality** always holds if $\mathbf{b} \in \langle A_1, A_2, \dots A_n \rangle$, where A_i denotes the i-th column of A.

Remark

Strong duality holds even when $\mathbf{b} \notin \langle A_1, A_2, \dots A_n \rangle$, that is there exists $\mathbf{z} \in \mathbb{R}^p$ such that

$$A^\top \mathbf{z} = \mathbf{0} \text{ and } \mathbf{b}^\top \mathbf{z} \neq \mathbf{0}.$$

It follows, indeed, that $\theta(\boldsymbol{\lambda})$ is also unbounded above along $\{t\mathbf{z} \,|\, t \in \mathbb{R}\}$.

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Exercise 1

Determine the maximum of the function

$$\begin{aligned} \max f(x,y) &= -7x^2 - y^2 \\ \text{s.t.} \quad & x \geq 0 \\ & y + 1 \geq 0 \end{aligned}$$

applying the KKT conditions.

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applying the KKT conditions.

Solution:

- First, the constraints should be written as:

$$\begin{aligned} g_1(x,y) &= -x \leq 0 \\ g_2(x,y) &= -y - 1 \leq 0, \end{aligned}$$

  given that the problem is a maximization problem.

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Exercise 1

- To evaluate if the objective function is **concave**, we compute the gradient and the Hessian matrix of f:

$$\nabla f(x,y) = \begin{bmatrix} -14x \\ -2y \end{bmatrix} \qquad H_f(x,y) = \begin{bmatrix} -14 & 0 \\ 0 & -2 \end{bmatrix}.$$

  Since $h_{11} = -14 < 0$ and $\det H_f = 28 > 0$, it follows that H_f is **negative definite**, therefore f is **concave**.

- We further notice that the constraints are *affine* functions, therefore **convex**. The results discussed up to now ensure that the KKT conditions are **sufficient** conditions.

- Then, we write the Lagrangian function as

$$\mathcal{L}(x,y,\lambda_1,\lambda_2) = -7x^2 - y^2 - \lambda_1(-x) - \lambda_2(-1-y) = -7x^2 - y^2 + \lambda_1 x + \lambda_2 + \lambda_2 y$$

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Exercise 1

- The KKT conditions are as follows:

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x} = 0 \\ \frac{\partial \mathcal{L}}{\partial y} = 0 \\ -x \leq 0 \\ \lambda_1 x = 0, \quad \lambda_1 \geq 0 \\ -y - 1 \leq 0 \\ \lambda_2(-y-1) = 0, \quad \lambda_2 \geq 0 \end{cases} \implies \begin{cases} \lambda_1 = 14x \\ \lambda_2 = 2y \\ x \geq 0 \\ \lambda_1 x = 0, \quad \lambda_1 \geq 0 \\ y \geq -1 \\ \lambda_2(-y-1) = 0, \quad \lambda_2 \geq 0 \end{cases}$$

- Starting from the complementarity conditions

$$\begin{cases} \lambda_1 x = 0, \quad \lambda_1 \geq 0 \\ \lambda_2(-y-1) = 0 \quad \lambda_2 \geq 0 \end{cases},$$

we derive the following cases:

a) $\lambda_1 x = 0 \iff \lambda_1 = 0 \vee x = 0$ b) $\lambda_2(-y-1) = 0 \iff \lambda_2 = 0 \vee y = -1.$

- Combining these possibilities, we get that

$$x = 0, \quad y = 0, \quad \lambda_1 = 0, \quad \lambda_2 = 0.$$

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- Combining these possibilities, we get that

$$x = 0, \quad y = 0, \quad \lambda_1 = 0, \quad \lambda_2 = 0.$$

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Exercise 1

- If we suppose that $y = -1$, then $\lambda_2 = -2$ which is not compatible with the condition $\lambda_2 \geq 0$. Therefore, $(0, 0)$ is the only solution.

- In $(0, 0)$ there is only one active constraint, then **the gradient of this constraint is linearly independent**.

- The active constraints are qualified, hence (0,0) is the **global maximum**.

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Exercise 2: Determine the maximum and the minimum absolute points of the function

$$\begin{aligned} f(x,y) &= x^2 + 2y^2 \\ \text{s.t.} \quad & x^2 + y^2 \leq 1 \\ & x + y - 1 = 0 \end{aligned}$$

applying the KKT conditions.

Exercise 3: Determine the minimum of the function

$$\begin{aligned} \min f(x,y) &= -x - y^2 \\ \text{s.t.} \quad & 2x^2 + y^2 \leq 7 \\ & x^2 + y^2 \geq 2 \end{aligned}$$

applying the KKT conditions.

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References

- Sébastien Bubeck. **Convex Optimization: Algorithms and Complexity**. Foundations and Trends in Machine Learning, 8(3-4):231 - 357, 2015.
- Stephen Boyd and Lieven Vandenberghe. **Convex Optimization**. Cambridge University Press, New York, NY, USA, 2004.

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